

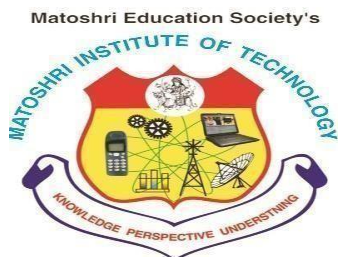
Matoshri Education Society's
MIT POLYTECHNIC AND ENGINEERING
Yeola

Vision

"Transforming rural students into competent engineering profession by providing academic excellence through value based and skill-oriented education."

Mission

- M1.** To prepare the students for attaining the knowledge of engineering fields.
- M2.** To regulate the faculty into modern technology establishing the liaison with industry.
- M3.** To prepare students for successful employment in industry, self-enterprise and higher education by using modern engineering tools with supportive faculties.
- M4.** To develop responsible citizen through awareness and Ethical value's



Matoshri Education Society's
MIT POLYTECHNIC AND ENGINEERING
Yeola

DEPARTMENT OF ELECTRICAL ENGINEERING

Vision

"To Impart Relevant value based technical education for the rural student to prepare effective, socially responsible engineer for electrical industry."

Mission

- M1.** To regulate the faculty into modern technology establishing the liaison with industry.
- M2.** To produce skilled technical manpower for the industry & technology
- M3.** To Instill strong ethical practices & value's
- M4.** To motivate the student for higher education& entrepreneurship



Matoshri Education Society's
MIT POLYTECHNIC AND ENGINEERING
YEOLA (NASHIK)



A

Project Report

On

“Touchless Doorbell System”

Submitted by-

Mr. Gomladu Rutik Santosh

Mr. Gaikwad Pavan Prashant

Mr. Bhad Kiran Sachin

Under Guidance of

Miss. BELDAR.G. N

DEPARTMENT OF ELECTRICAL ENGINEERING
MATOSHRI INSTITUTE OF TECHNOLOGY,
YEOLA (NASHIK)

2025-2026

MATOSHRI EDUCATION SOCIETY'S
MIT POLYTECHNIC AND ENGINEERING
YEOLA (NASHIK)
ANNEXURE A



CERTIFICATE



This is to certify that, following students have successfully
complete their project on

“Touchless Doorbell System”

Mr. Gomladu Rutik Santosh

Mr. Gaikwad Pavan Prashant

Mr. Bhad Kiran Sachin

From **MIT POLYTECHNIC AND ENGINEERING, Dhanore, Yeola** has
Completed *Project of second year* having title **“Touchless Doorbell System”** during
the academic year 2025-2026. The project Completed in group of consisting of
Three persons under the guidance of the **Miss. Beldar.G. N**

Date:

Place: Yeola

Miss. Beldar.G.N
Guide

Mr. Gujar N.P.
H.O.D

External Examiner

Mr. Gujarathi G.S.
Principal

ACKNOWLEDGEMENT

I feel Great pleasure to present the project entitled “**Touchless Doorbell System**” But it would be unfair on our part if I do not acknowledge efforts of some of the people without the support of whom, this project would not have been a success.

First and for most I am very much thankful to my respected Guide Miss. Beldar.G.N.for his leading guidance in this project. Also, he has been persistent source of inspiration to me. I would like to express my sincere thanks and appreciation to HOD **Mr. Gujar N.P.** Principal **Mr. Gujarathi G.S.** & project coordinator Miss. Beldar.G.N for his/her valuable support.

Most importantly I would like to express our sincere gratitude towards me **Friends & Family** for always being there when I needed them most.

Mr. Gomladu Rutik Santosh
Mr. Gaikwad Pavan Prashant
Mr. Bhad Kiran Sachin

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Abstract:

The Touchless Doorbell is an innovative system designed to enhance hygiene and convenience by eliminating the need for physical contact when ringing a doorbell.

Traditional doorbells require touch, which can transmit germs and viruses, especially in public or shared spaces. This system uses infrared (IR) or ultrasonic sensors to detect a person's hand motion in front of the doorbell. When motion is detected within a certain range, the system automatically triggers the bell sound. It operates efficiently without requiring physical pressure or contact.

The sensor converts hand movement into an electrical signal to activate the buzzer or chime. A microcontroller processes the signal and controls the sound system accordingly.

The design is user-friendly, cost-effective, and easy to install. It can be powered using batteries, USB, or a small power adapter. The system also reduces maintenance because there are no moving mechanical parts. It is especially useful in hospitals, offices, schools, and homes where hygiene is a priority.

The project promotes a touch-free interaction experience, aligning with modern smart home technology trends. It can also be integrated with wireless modules for remote operation or notification. The use of simple sensors makes it affordable and energy-efficient.

Its response time is quick, ensuring immediate ringing without delay. The circuit design is compact and reliable for continuous operation.

The prototype demonstrates the working principle effectively in real-world conditions. Future upgrades can include camera integration or Iota connectivity for enhanced functionality. Overall, the Touchless Doorbell provides a safe, smart, and modern alternative to conventional doorbells.

Introduction:

The Touchless Doorbell is a modern and hygienic solution designed to replace conventional push-button doorbells. In traditional systems, visitors must physically press the switch to ring the bell, which can lead to the spread of germs and viruses.

The global pandemic has increased awareness about touch-free technologies that promote better hygiene. This project aims to create a safe and convenient alternative that operates without any physical contact. The system works using sensors such as infrared (IR) or ultrasonic detectors to sense motion or hand gestures.

When a person waves their hand near the sensor, it automatically triggers the ringing mechanism. This eliminates the need for any physical touch and reduces the risk of contamination. The system is simple, reliable, and easy to install in both homes and public buildings. It operates efficiently with low power consumption.

The design includes key components such as sensors, a microcontroller, a power supply, and a buzzer. The microcontroller processes the signal from the sensor and activates the sound circuit. The device can be powered using batteries or a small DC adapter. It can also be integrated with wireless systems for smart home applications.

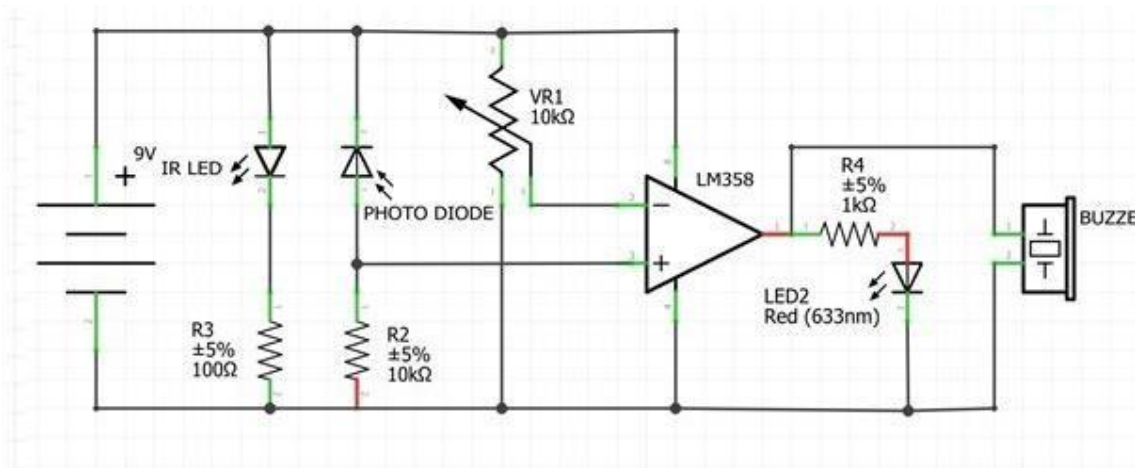
The touchless design adds a modern look to doors and entrances. It is ideal for hospitals, offices, restaurants, and residential buildings. The system ensures quick response and long-term durability because there are no mechanical parts. In the future, the system can be enhanced with camera or mobile app integration. Thus, the Touchless Doorbell represents a small but impactful step toward smarter and safer living environments.

Principle of Touchless Doorbell:

The Touchless Doorbell works on the principle of motion detection using infrared (IR) or ultrasonic sensing technology. When a person brings their hand near the sensor, it detects the presence or movement by measuring the change in reflected light or sound waves. In an IR-based system, an infrared LED emits light that gets reflected back when an object, such as a hand, comes close. The reflected signal is received by an IR receiver, which then generates an electrical signal. In an ultrasonic-based system, the sensor sends high-frequency sound waves and measures the time taken for the echo to return after striking an object.

The change in distance or reflection time indicates motion, which triggers the circuit. This signal is then sent to a microcontroller or transistor-based circuit that processes the signal and activates a buzzer or chime. Thus, without any physical contact, the doorbell rings automatically when motion is detected near the sensor. The principle relies on the conversion of motion or proximity into an electrical signal using sensors and electronic switching. This ensures contactless operation, providing enhanced hygiene, safety, and convenience.

Circuit Diagram:



Working:

A touchless doorbell works without pressing any physical button. It uses a proximity sensor to detect when someone is near the door. The sensor is placed at the entrance of the house.

When a person brings their hand close to the sensor, it senses the object. The sensor sends out signals, such as infrared light or ultrasonic waves.

These signals travel until they hit an object. When the signals bounce back, the sensor receives the reflected signal. The sensor then detects the presence of the object. This detection sends an electrical signal to the control circuit.

The control circuit is connected to a buzzer or speaker. When the circuit receives the signal, it activates the buzzer. The buzzer produces a ringing sound to alert the people inside. Once the hand is removed, the sensor stops detecting motion.

The control circuit then stops the signal to the buzzer. The ringing stops automatically. The system works very quickly, usually in less than a second. It is powered by a battery or DC adapter.

Some models have an LED light that glows when the sensor is activated. The sensor's detection range can be adjusted for accuracy. No physical contact is needed, making the system hygienic and safe. It is useful in homes, offices, and hospitals. The components are compact and easy to install. The system consumes very little power. It provides a modern and convenient alternative to traditional doorbells. Overall, a touchless doorbell is safe, hygienic, and easy to use.

Components Used in Touchless Doorbell using Proximity Sensor:

1) 9v Battery:



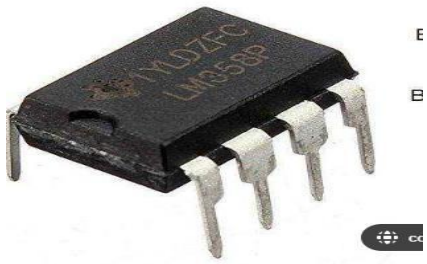
The replaceable battery has the highest energy density, flattest discharge voltage curve, longest shelf life, widest operating temperature range, and lightest weight of any comparable 9-volt battery. The battery meets ANSI Alkaline 1604 size specifications and has overcurrent protection.

2) 10k Variable Resistor



10K ohm Preset Cermet [103] [RM065] Variable Resistor Cermet preset is a compact variable resistor and PCB mountable with 3 terminal pins. The voltage between the terminal varies as the preset is rotated. The Variable resistors are used for varying voltage as per the need in a circuit.

3) LM358



The LM358 is a low-power dual operational amplifier (op-amp) IC containing two independent, high-gain op-amps in a single 8-pin package, designed to operate from a single power supply. It features a wide supply voltage range (3V to 32V), a low supply current, and an input common-mode voltage range that includes ground, making it suitable for battery-powered circuits and applications like sensor amplification, active filters, and comparator circuits.

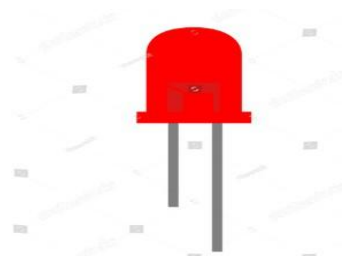
4) Buzzer



A buzzer is an electrical device that generates an audible, buzzing or beeping sound by using vibrations in a diaphragm or metal plate when supplied with an external voltage.

It functions as an audio signaling device, acting as a transducer to convert electrical energy into sound waves. Buzzers are powered by direct current (DC) voltage and are commonly used in products like printers, alarm clocks, computers, and electronic toys for various alarm and confirmation functions.

5) LED



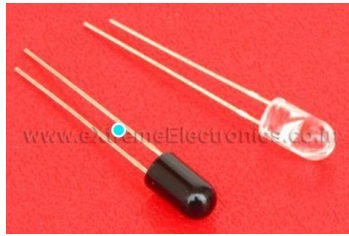
A red LED is a type of Light-Emitting Diode that produces red light, a specific wavelength of the visible light spectrum with the longest wavelength. These diodes are used in various applications, such as status indicators on electronic devices and for red light therapy, a non-invasive treatment that uses low-level red and near-infrared light to stimulate cellular function, potentially benefiting skin rejuvenation, hair growth, and wound healing.

6) Zero PCB



A Zero PCB, also known as a preboard, Dot PCB, or general-purpose PCB, is a bare, pre-drilled board with a grid of holes for manual prototyping of electronic circuits. It features isolated copper pads around each hole, allowing users to manually connect components with wires and solder to create custom circuits without a predefined layout.

7) IR RX TX



IR TX (transmitter) is an infrared light-emitting diode (LED) that emits infrared radiation (invisible to the human eye) to send signals, while IR RX (receiver) is an infrared photodiode that detects this emitted radiation. When paired, the IR TX and RX work together in systems like remote controls and sensors to send and receive coded information, often on a specific wavelength (e.g., 940nm).

8) Resistor



A resistor is an electronic component that resists or impedes the flow of electric current in a circuit, measured in ohms (Ω). It's a passive component that limits current, divides voltage, and dissipates energy as heat, making it essential for controlling and regulating electricity in virtually all electronic devices. Resistors come in various types, sizes, and materials, and are represented in circuit diagrams by a zigzag line (ANSI) or a rectangle (IEC).

Advantages Of Touchless Doorbell:

1. Hygienic and Safe:

No physical contact is needed, reducing the spread of germs and viruses.

2. Convenient Operation:

Visitors can ring the bell just by waving their hand —quick and easy to use.

3. Energy Efficient:

Many touchless doorbells use low power or motion sensors that consume very little energy.

4. Modern and Smart Look:

Gives a stylish, modern appearance to homes or offices.

5. Integration with Smart Devices:

Can be connected to smartphones or smart home systems for notifications and video streaming.

6. Weather Resistant:

Since there are no buttons to press, it works well even in rainy or dusty conditions.

7. Low Maintenance:

Fewer moving parts mean less wear and tear over time.

8. Customizable Features:

Some models allow voice, gesture, or proximity settings for personalized use.

Disadvantage of Touchless Doorbell:

1. Higher Cost:

More expensive than traditional doorbells due to sensor technology.

2. Power Dependency:

Requires continuous power or batteries; may not work during power cuts if not backed up.

3. False Triggers:

Can get activated by wind, pets, or passing objects if sensitivity is high.

4. Limited Range:

Sensor may not detect hand motion properly if the distance or angle is off.

5. Complex Installation:

Installation and setup may require technical knowledge or professional help.

6. Maintenance of Sensors:

Sensors may get affected by dust, sunlight, or rain, reducing performance over time.

Application of Touchless Doorbell:

1. Residential Homes:

Used at the main door to provide hygienic and convenient visitor alerts.

2. Offices and Commercial Buildings:

Ensures contactless entry for employees and visitors.

3. Hospitals and Clinics:

Helps maintain hygiene and prevent infection spread.

4. Schools and Colleges:

Used at main gates or staff rooms for touch-free operation.

5. Hotels and Restaurants:

Enhances customer experience with modern, hygienic door systems.

6. Industrial Areas:

Used where workers have dirty or gloved hands, allowing contactless use.

Conclusion:

A touchless doorbell is a modern, hygienic, and convenient alternative to traditional doorbells. It helps reduce physical contact, prevents the spread of germs, and provides a smart and stylish solution for homes and public places. Although it may be costlier and require careful installation, its benefits in safety and ease of use make it an ideal choice for today's smart and health-conscious environments.